

1. A CSTR reactor has to be designed for the following third order, liquid phase reaction:

$3A \rightarrow 3B + C$ ($k = 0.2 \text{ L}^2/\text{mol}^2 \cdot \text{s}$). Stoichiometric feed of A and B enters a reactor along with an inert I. The total inlet molar flow rate of 2400 mol/min. The mole fraction of A in the inlet, $x_{A0} = 0.75$, and v_0 is 100 L/min.

a) Make the stoichiometric table for the species.

Species	initial	change	remaining
A	1800	-1800x	1800(1-x)
B	0	+1800x	1800x
C	0	+600x	600x
I	F_{I0}	0	$F_I = F_{I0}$

b) Find the CSTR volume required to achieve a conversion of 0.8.

$$X = 0.8$$

$$\text{In-Out} + G_{\text{gen}}^0 - \text{Cons} = \text{Accum}^0$$

$$F_{A0} - F_A - r_A V = 0$$

$$r_A V = F_{A0} - F_A$$

$$V = \frac{F_{A0} - F_A}{r_A} = \frac{F_{A0} - F_A}{k C_A^3}$$

$$F_{A0} = 1800 \frac{\text{mol}}{\text{min}}$$

$$F_A = 1800(1-0.8) = 360 \frac{\text{mol}}{\text{min}}$$

$$r_A = k C_A^3$$

$$C_{A0} = \frac{F_{A0}}{v_0} = \frac{1800 \frac{\text{mol}}{\text{min}}}{100 \frac{\text{L}}{\text{min}}} = 18 \frac{\text{mol}}{\text{L}}$$

$$C_A = 18 \frac{\text{mol}}{\text{L}} (1-0.8) = 3.6 \frac{\text{mol}}{\text{L}}$$

$$V = \frac{1800 \frac{\text{mol}}{\text{min}} - 360 \frac{\text{mol}}{\text{min}}}{0.2 \frac{\text{L}^2}{\text{mol}^2 \cdot \text{s}} \cdot \frac{60 \text{ s}}{1 \text{ min}} \cdot \left[3.6 \frac{\text{mol}}{\text{L}}\right]^3}$$

$$V = 2.57 \text{ L}$$

2. A reversible, gas phase reaction ($A + B \rightleftharpoons C + D$) is taking place in a PFR. The inlet concentration of A and B are 3 moles/L, $v_0 = 0.5$ L/s, with no C or D in the feed. The forward reaction rate constant is $3 \text{ mol}/(\text{m}^3 \cdot \text{atm}^2 \cdot \text{s})$ and the equilibrium constant is 1. If the total pressure is 3 atm:

a) What is the maximum conversion possible in this reaction?

	A	+	B	$\rightleftharpoons$	C	+	D
Initial	1.0		1.0		0		0
Change	-0.5		-0.5		+0.5		+0.5
Remaining	0.5		0.5		0.5		0.5

Also $\rightarrow$

max conversion: $r_A = 0$

$$P_A = P_B$$

$$P_{A0}(1-x) = P_{A0}x$$

$$1-x = x$$

$$2x = 1$$

$$x = 0.5$$

Because the reaction proceeds in both the forward and reverse direction, the maximum conversion is $x = 0.5$. Any other value would cause the reaction to shift directions.

b) Find an expression for conversion V vs volume of the reactor? For what volume of the reactor will the conversion be 0.42?

$$V = f(x)$$

$$\frac{dF_A}{dV} = r_A$$

$$\int_0^V dV = \int_0^{F_{A0}} \frac{dF_A}{r_A}$$

$$\int_0^V dV = \int_0^x \frac{-F_{A0}}{r_A} dx \quad [dF_A = -F_{A0} dx]$$

$$\int_0^V dV = \int_0^x \frac{-F_{A0}}{-k P_{A0}^2 (1-2x)} dx$$

$$\int_0^V dV = \frac{F_{A0}}{k P_{A0}^2} \int_0^x \frac{1}{(1-2x)} dx$$

$$V = \frac{F_{A0}}{k P_{A0}^2} \left[-\frac{1}{2} \ln(1-2x) \right]_{x=0}^{x=x}$$

[if integrating from initial conditions then bounds are V_0 to V and x_0 to x]

$$V = \frac{-F_{A0}}{2 k P_{A0}^2} \left[\ln(1-2x) \right]_{x=0}^{x=0.42} = \frac{-1.5 \frac{\text{mol}}{\text{s}}}{2 \cdot \left[3 \frac{\text{mol}}{\text{m}^3 \cdot \text{atm}^2 \cdot \text{s}} \right] \left[1.5 \text{ atm} \right]^2} \ln(1-2(0.42))$$

$$V = 0.204 \text{ m}^3$$

$$r_A = -k_1 P_A P_B + k_{-1} P_C P_D$$

$$r_A = -k P_A^2 + k P_B^2 \quad [k_1 = k_{-1}, P_A = P_B, P_C = P_D]$$

$$r_A = -k [P_{A0}(1-x)]^2 + k [P_{A0}x]^2$$

$$r_A = -k P_{A0}^2 (1-2x)$$

$$P_{A0} = 1.5 \text{ atm}$$